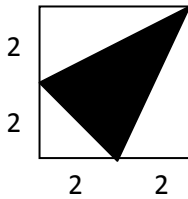


Q1	If $A \diamond B = AB^3$ what is the value of $(2 \diamond 3) \diamond 2$?	Q2	What is the SUM of the highest common factor (HCF) and lowest common multiple (LCM) of the numbers 100 and 24 ?
	5		5
Q3	Find the average of the four prime numbers that are closest to 20.	Q4	In cooking 4 tablespoons = $\frac{1}{4}$ cup 1 cup = 250 ml. How many tablespoons are there in a LITRE?
	5		5
Q5		$2 \clubsuit 4 = 20$ $3 \clubsuit 5 = 34$ $6 \clubsuit 9 = 117$ so $4 \clubsuit 5 = ?$	
		5	

Q6	50 apples = 20 pears 30 pears = 12 fish 100 fish = 1 dog so how many apples equal 1 dog?	Q7	I have an average of 78% for my first three tests in Mathematics. What mark must I get in my fourth test so that my overall average is 80%?
	5		5
Q8	$20 \div 14 = 1,428571428571....$ has a recurring string of decimal digits. What digit is in position 2014 to the right of the decimal comma?	Q9	The height of a rectangle is <u>increased</u> by 20% and the base is <u>decreased</u> by 20%. By what % is the area of the original rectangle decreased?
	5		5
	Q10	B = ?	
		5	

Q11

A triangle is formed in a square.



Find the area of the triangle.

10**Q12**

If the arrangement of numbers shown below continues,

what number would be directly below 144 ?

			1			
		2	3	4		
	5	6	7	8	9	
10	11	12	13	14	15	16

10**Q13**

The mathematical operation @ is defined by

$$a @ b = b^2 \div a$$

If $(a @ 2) @ 6 = 72$
then $a =$

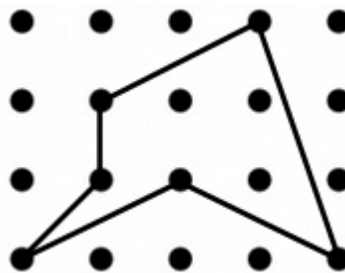
10**Q14**

A Pythagorean quadruple satisfies

$$a^2 + b^2 + c^2 = d^2$$

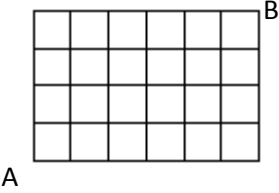
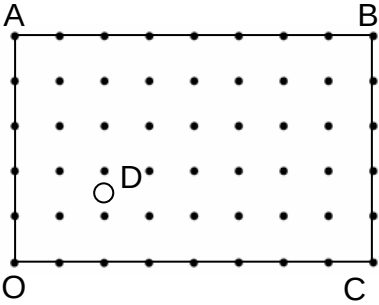
(a,b,c,d are natural numbers)

If three of the numbers are 272, 306 and 578, what is the fourth number?

10**Q15**

The dots are equally spaced.
What is the area of the (irregular) hexagon?

10

Q16	A bag of jellybeans contains 335 red 357 blue 592 pink 294 orange 520 white What is the <u>minimum</u> number of jellybeans you must pick to guarantee at least one bean of each colour?	Q17	You can pick any 3 digits from the set 2 5 5 8 8 How many different 3-digit numbers can you make?
	20		20
Q18	Starting at A, and finishing at B, how many different pathways are there if you can only move up or to the right? 	Q19	A fair 6-sided die is rolled twice. What is the probability that the outcome of the second roll exceeds the outcome of the first roll? <u>WRITE YOUR ANSWER AS A SIMPLIFIED FRACTION</u>
	20		20
Q20	A snooker ball is positioned at point D with co-ordinates (2 ; 1,5). Regard O as the origin. The ball is to be hit against side AB and must go straight to point C. What is the <u>x co-ordinate</u> of the point on AB that D must “hit” to ensure the ball goes directly to C? 		
			20

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SENIOR (2014) [Grade 10 & 11]

Q1 432	Q2 604	Q3 18	Q4 64	Q5 41	5 each
Q6 625	Q7 86	Q8 5	Q9 4	Q10 31	5 each
Q11 6	Q12 168	Q13 8	Q14 408	Q15 5	10 each
Q16 1805	Q17 18	Q18 210	Q19 5/12 (must be simplified fraction)	Q20 4,47 (must be 2 dp)	20 each

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SENIOR SECTION (Grade 8 & 9)



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SCORE

/200

Q1	Q2	Q3	Q4	Q5	5 marks each
Q6	Q7	Q8	Q9	Q10	5 marks each
Q11	Q12	Q13	Q14	Q15	10 marks each
Q16	Q17	Q18	Q19	Q20	20 marks each